

$$\begin{aligned} & \text{HL}^{11-12} \rightarrow \frac{1}{16\pi^2} \left(\frac{1}{144} \frac{1}{m_e^2} S_Y^2 Y_e^{i1p} (7 g_1^2 \overline{Y_e}^{i2p} - 12 c_Y^2 \overline{Y_e}^{rp} \overline{Y_e}^{i2s} Y_e^{rs}) - \right. \\ & \frac{1}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_d^p] \delta_{i1i2} + \frac{5}{72} \sum_p g_1^4 \text{LF}_{4,-1} [m_d^p] \delta_{i1i2} - \frac{4}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_d^p] \delta_{i1i2} - \\ & \frac{1}{9} \sum_p g_1^4 \text{LF}_{3,0} [m_e^p] \delta_{i1i2} + \frac{5}{24} \sum_p g_1^4 \text{LF}_{4,-1} [m_e^p] \delta_{i1i2} - \frac{4}{45} \sum_p g_1^4 \text{LF}_{5,-2} [m_e^p] \delta_{i1i2} - \\ & \frac{1}{18} \sum_p g_1^4 \text{LF}_{3,0} [m_l^p] \delta_{i1i2} + \frac{5}{48} \sum_p g_1^4 \text{LF}_{4,-1} [m_l^p] \delta_{i1i2} - \frac{2}{45} \sum_p g_1^4 \text{LF}_{5,-2} [m_l^p] \delta_{i1i2} - \\ & \frac{1}{54} \sum_p g_1^4 \text{LF}_{3,0} [m_q^p] \delta_{i1i2} + \frac{5}{144} \sum_p g_1^4 \text{LF}_{4,-1} [m_q^p] \delta_{i1i2} - \frac{2}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_q^p] \delta_{i1i2} - \\ & \frac{4}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_u^p] \delta_{i1i2} + \frac{5}{18} \sum_p g_1^4 \text{LF}_{4,-1} [m_u^p] \delta_{i1i2} - \frac{16}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_u^p] \delta_{i1i2} + \\ & \frac{1}{6} S_Y^2 Y_e^{i1p} (g_1^2 \overline{Y_e}^{i2p} - 3 c_Y^2 \overline{Y_e}^{rp} \overline{Y_e}^{i2s} Y_e^{rs}) \text{LF}_{1,2} [m_\Phi] - \frac{1}{24} g_1^2 S_Y^2 \overline{Y_e}^{i2p} Y_e^{i1p} \text{LF}_{2,1} [m_\Phi] + \\ & \frac{1}{72} (3 g_1^2 S_Y^2 \overline{Y_e}^{i2p} Y_e^{i1p} - 4 g_1^4 \delta_{i1i2}) \text{LF}_{3,0} [m_\Phi] + \frac{5}{48} g_1^4 \text{LF}_{4,-1} [m_\Phi] \delta_{i1i2} - \\ & \frac{2}{45} g_1^4 \text{LF}_{5,-2} [m_\Phi] \delta_{i1i2} - \frac{1}{18} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} + \frac{4}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{2} g_1^2 c_Y^2 \overline{Y_e}^{i2p} Y_e^{i1p} \text{LF}_{2,1,0} [m_1, m_e^p] - g_1^2 c_Y^2 \overline{Y_e}^{i2p} Y_e^{i1p} \text{LF}_{3,1,-1} [m_1, m_e^p] + \\ & \frac{1}{2} g_1^2 c_Y^2 \overline{Y_e}^{i2p} Y_e^{i1p} \text{LF}_{4,1,-2} [m_1, m_e^p] - \frac{1}{96} g_1^4 \text{LF}_{2,1,0} [m_1, m_l^{i1}] \delta_{i1i2} - \\ & \frac{1}{96} g_1^4 \text{LF}_{2,2,-1} [m_1, m_l^{i1}] \delta_{i1i2} + \frac{1}{48} g_1^4 \text{LF}_{3,1,-1} [m_1, m_l^{i1}] \delta_{i1i2} - \\ & \frac{1}{96} g_1^4 \text{LF}_{4,1,-2} [m_1, m_l^{i1}] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{2,1,0} [m_1, m_l^{i2}] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{2,2,-1} [m_1, m_l^{i2}] \delta_{i1i2} + \\ & \frac{1}{48} g_1^4 \text{LF}_{3,1,-1} [m_1, m_l^{i2}] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{4,1,-2} [m_1, m_l^{i2}] \delta_{i1i2} - \\ & \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_l^{i1}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,2,-1} [m_2, m_l^{i1}] \delta_{i1i2} + \\ & \frac{1}{16} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_l^{i1}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{4,1,-2} [m_2, m_l^{i1}] \delta_{i1i2} - \\ & \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_l^{i2}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,2,-1} [m_2, m_l^{i2}] \delta_{i1i2} + \\ & \frac{1}{16} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_l^{i2}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{4,1,-2} [m_2, m_l^{i2}] \delta_{i1i2} - \\ & \frac{1}{6} g_1^2 (c_Y \overline{a_d}^{pr} - s_Y \tilde{\mu} \overline{y_d}^{pr}) (c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr}) \text{LF}_{2,2,0} [m_d^r, m_q^p] \delta_{i1i2} + \\ & \frac{1}{12} g_1^2 (c_Y \overline{a_d}^{pr} - s_Y \tilde{\mu} \overline{y_d}^{pr}) (c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr}) \text{LF}_{3,2,-1} [m_d^r, m_q^p] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 \overline{Y_e}^{i2p} Y_e^{i1p} \text{LF}_{2,1,0} [m_e^p, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{Y_e}^{i2p} Y_e^{i1p} \text{LF}_{2,2,-1} [m_e^p, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{Y_e}^{i2p} Y_e^{i1p} \\ & \text{LF}_{3,1,-1} [m_e^p, \tilde{\mu}] - \frac{1}{6} g_1^2 (c_Y \overline{a_e}^{pr} - s_Y \tilde{\mu} \overline{y_e}^{pr}) (c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr}) \text{LF}_{2,2,0} [m_e^r, m_l^p] \delta_{i1i2} + \\ & \frac{1}{12} g_1^2 (c_Y \overline{a_e}^{pr} - s_Y \tilde{\mu} \overline{y_e}^{pr}) (c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr}) \text{LF}_{3,2,-1} [m_e^r, m_l^p] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_Y \overline{a_e}^{pr} - s_Y \tilde{\mu} \overline{y_e}^{pr}) (c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr}) \text{LF}_{3,1,0} [m_l^p, m_e^r] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_Y \overline{a_e}^{pr} - s_Y \tilde{\mu} \overline{y_e}^{pr}) (c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr}) \text{LF}_{3,2,-1} [m_l^p, m_e^r] \delta_{i1i2} - \\ & \frac{3}{8} g_1^2 (c_Y \overline{a_e}^{pr} - s_Y \tilde{\mu} \overline{y_e}^{pr}) (c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr}) \text{LF}_{4,1,-1} [m_l^p, m_e^r] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_Y \overline{a_e}^{pr} - s_Y \tilde{\mu} \overline{y_e}^{pr}) (c_Y a_e^{pr} - s_Y \tilde{\mu} y_e^{pr}) \text{LF}_{5,1,-2} [m_l^p, m_e^r] \delta_{i1i2} + \\ & \frac{1}{48} g_1^4 \text{LF}_{2,1,0} [m_l^{i1}, m_1] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{3,1,-1} [m_l^{i1}, m_1] \delta_{i1i2} + \\ & \frac{1}{16} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_l^{i1}, m_2] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_l^{i1}, m_2] \delta_{i1i2} + \\ & \frac{1}{48} g_1^4 \text{LF}_{2,1,0} [m_l^{i2}, m_1] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{3,1,-1} [m_l^{i2}, m_1] \delta_{i1i2} + \\ & \frac{1}{16} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_l^{i2}, m_2] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_l^{i2}, m_2] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_Y \overline{a_d}^{pr} - s_Y \tilde{\mu} \overline{y_d}^{pr}) (c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr}) \text{LF}_{3,1,0} [m_q^p, m_d^r] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_Y \overline{a_d}^{pr} - s_Y \tilde{\mu} \overline{y_d}^{pr}) (c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr}) \text{LF}_{3,2,-1} [m_q^p, m_d^r] \delta_{i1i2} - \\ & \frac{5}{8} g_1^2 (c_Y \overline{a_d}^{pr} - s_Y \tilde{\mu} \overline{y_d}^{pr}) (c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr}) \text{LF}_{4,1,-1} [m_q^p, m_d^r] \delta_{i1i2} + \\ & \frac{1}{2} g_1^2 (c_Y \overline{a_d}^{pr} - s_Y \tilde{\mu} \overline{y_d}^{pr}) (c_Y a_d^{pr} - s_Y \tilde{\mu} y_d^{pr}) \text{LF}_{5,1,-2} [m_q^p, m_d^r] \delta_{i1i2} + \\ & \frac{1}{12} g_1^2 (s_Y \overline{a_u}^{pr} - \tilde{\mu} c_Y \overline{y_u}^{pr}) (s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr}) \text{LF}_{2,2,0} [m_q^p, m_u^r] \delta_{i1i2} - \\ & \frac{1}{24} g_1^2 (s_Y \overline{a_u}^{pr} - \tilde{\mu} c_Y \overline{y_u}^{pr}) (s_Y a_u^{pr} - \tilde{\mu} c_Y y_u^{pr}) \text{LF}_{3,2,-1} [m_q^p, m_u^r] \delta_{i1i2} - \\ & \frac{1}{12} g_1^2 (s_Y \$$